

Bath Planner: the AI Bathroom Designer & Planner

THE PROBLEM

- **Four constraints, no single tool**
Room dimensions, code clearances, product compatibility, and budget all have to line up at once.
- **Showrooms sell options, not plans**
A catalog shows what exists, not what actually fits an exact 5'×8' room.
- **Mistakes surface too late**
A fixture picked for looks can turn out not to clear the door swing, or blow the budget, only after it's already ordered.

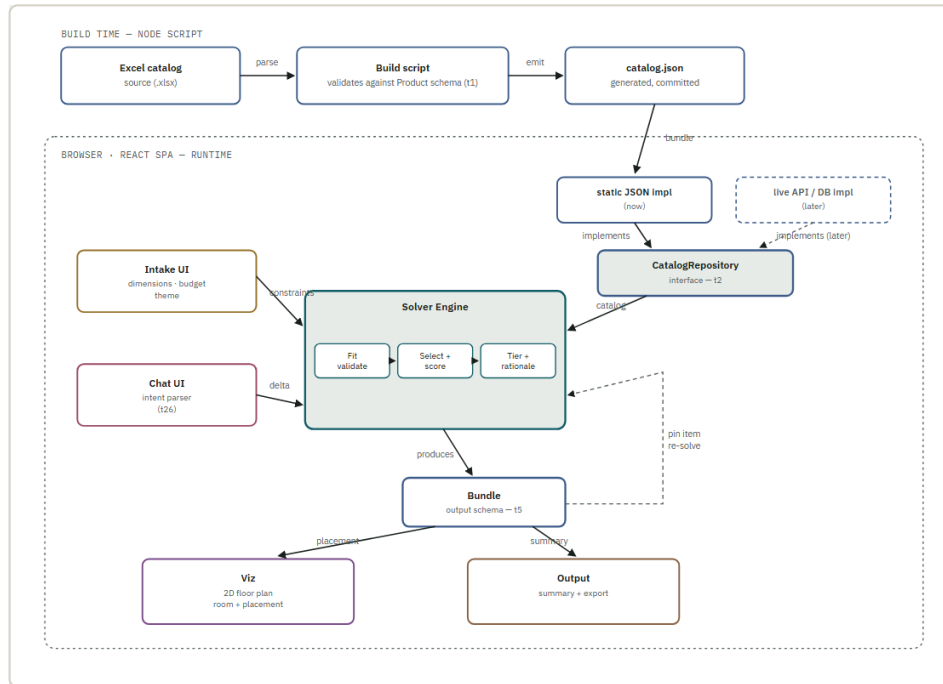
THE SOLUTION

- **A real constraint solver**
Takes room dimensions, a budget, and a style, and returns bundles already checked against real clearances.
- **Three ready-made tiers**
Value, Balanced, and Premium, each fully priced with a plain-language reason for every pick.
- **Refine it by talking to it**
“Swap the vanity for something cheaper” re-solves live and explains what changed and why.

Live: kohlerbathplanner.vercel.app
github.com/Alok49225/kohlerbathplanner

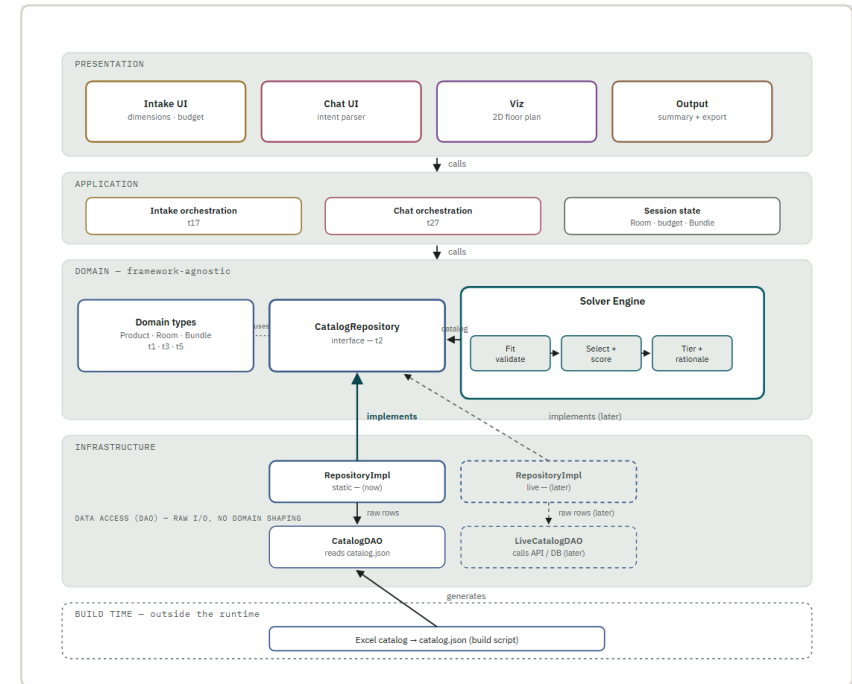
How the System Works

DATA FLOW



Excel → catalog.json at build time; Intake/Chat feed the Solver Engine, which produces a Bundle for Viz + Output.

LAYERS & DEPENDENCY DIRECTION

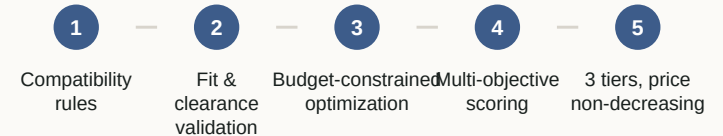


Every arrow points down except one: Infrastructure implements an interface Domain owns.

BUILT AI-ASSISTED, SPEC-FIRST

All 35 build tasks were proposed as a concrete spec, approved, then implemented and tested by Claude before the next began. Every post-launch bug was reproduced and root-caused live in the running app — never guessed. See the Prompts Documentation for the full record.

THE SOLVER PIPELINE



Key Features

SHIPPED TODAY

- **Click-to-place plumbing**
Set rough-ins by clicking the floor plan; wall and clearance direction are inferred automatically.
- **3 auto-priced tiers**
Value, Balanced, and Premium bundles, price guaranteed non-decreasing across tiers.
- **Conversational refinement**
“Swap the vanity for something cheaper” re-solves live and explains the change.
- **Explainable rationale**
Every line item states why it was chosen, not just what it costs.
- **Water-efficiency scoring**
Every bundle carries a sustainability score, surfaced directly in the summary.

ON THE ROADMAP — NOT YET BUILT

AI-rendered room visualization

An AI image model generating a photo-realistic view of the solved bundle inside the actual room, clearly labeled as illustrative rather than a literal product photo — the “AI concept render” already scoped as a Could-have in the original blueprint, not a new idea invented for this deck.

Technology & Results

React + TS + Vite

Fully static, zero backend

310 / 310

automated tests passing

35 / 35

build tasks complete

Live

kohlerbathplanner.vercel.app

7

real bugs found & fixed
post-launch

THE WORKING MODEL — LIVE ON KOHLERBATHPLANNER.VERCEL.APP

swap the vanity for something cheaper

Swapped the vanity for Derwent 48in Double Vanity — The Derwent 48in Double Vanity was picked for its double-bowl look, matching your minimalist modern style..

Type a message...

Send

A real chat exchange — swap request, live re-solve, rationale

Custom

- value: \$3,250 (2 warnings)
- balanced: \$4,232
- premium: \$4,997

Value	Balanced	Premium
\$3,250	\$4,232	\$4,997

Toilet Vanity Faucet Shower Lighting

Item	Price
TOILET	\$1,899
Veil Wall-Hung Toilet - Kohler, white	
The Veil Wall-Hung Toilet was picked for its wall-hung look, matching your minimalist modern style.	
VANITY	\$2,450
Derwent 48in Double Vanity - Kohler, matte-black	
The Derwent 48in Double Vanity was picked for its double-bowl look, matching your minimalist modern style.	
FAUCET	\$189
Forte Single-Handle Faucet - Kohler, chrome	
The Forte Single-Handle Faucet is the most budget-friendly faucet that still fits your room.	
SHOWER	\$329
Forte Multifunction Shower Kit - Kohler, chrome	
The Forte Multifunction Shower Kit is the most budget-friendly shower that still fits your room.	
LIGHTING	\$129
Forte Flush Mount - Kohler, white	
The Forte Flush Mount is the most budget-friendly lighting that still fits your room.	
Total: \$4,996 of \$5,000 budget	\$4 remaining
Water efficiency: 69%	

Solved bundle — tiers, floor plan, explainable pricing